Benefits and costs of the ETS in the EU, a lesson learned for the CBAM design

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Any opinions and conclusions expressed in this discussion do not necessarily represent the views of the Banque de France.

This paper: What it does

- In the context of the EU "Fit for 55" package, it evaluates *ex post* (i) the efficiency of the ETS and (ii) its potential anti-competitive effects
- Takes these results as insights for the design of the envisaged CBAM. In particular:
  - Carbon leakage
  - Anti-competitive effects
- Combines different types of datasets to empirically study these questions:
  - Trade flows and production values (AMNE OECD's Input-Output tables)
  - CO_2 emissions (WIOD Environmental Accounts)
  - Regulated sectors (EU ETS data)
  - Emission prices (ICAP Allowance Price Explorer)

This paper: What it finds

- Performs OLS estimations and finds that an increase in ETS coverage is correlated with:
  - A reduction in GHG emissions by 2/2.5 percentage points per year
  - An increase in carbon leakage: intensification of emissions outside the EU
  - A decrease of domestic companies competitiveness, by reducing more their gross output in comparison to foreign companies

Comment #1

- If lagged increase in ETS stringency reduces emissions, it seems that contemporaneous increase in ETS stringency increases emissions, with a similar magnitude

  $\longrightarrow$ How can we rationalize this?

VARIABLES	(1) $log(em)$	(2) $log(em)$	(3) $\Delta log(em)$	(4) $\Delta log(em)$
$log(em)_{t-1}$	0.835***	0.813***	-0.0863***	-0.106***
	(0.00805)	(0.00825)	(0.00626)	(0.00670)
$sh.emtraded_t$	0.0298***	0.0291***	0.00631***	0.0124***
	(0.00192)	(0.00190)	(0.00159)	(0.000975)
$sh.emtraded_{t-1}$	-0.0203***	-0.0196***	-0.000874	-0.00963***
	(0.00173)	(0.00171)	(0.00137)	(0.000976)
$ETS3 : sh.emtraded_t$	-0.000970	0.00507**	0.0126***	
	(0.00220)	(0.00233)	(0.00175)	
$ETS3 : sh.emtraded_{t-1}$	-0.00368*	-0.00517**	-0.0171***	
	(0.00209)	(0.00212)	(0.00157)	
$ETS3 : p.em.sh._t$		-0.000172**		0.000239***
		(7.06e-05)		(6.25e-05)
$\Delta log(em)_{t-1}$			-0.282***	-0.286***
			(0.00953)	(0.00962)
$ETS3 : p.em.sh._{t-1}$				-0.000403***
				(7.27e-05)
Observations	5,408	5,408	4,941	4,941
Number of id	459	459	453	453
Country FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Industry Trend		YES		YES
R2 overall	0.889	0.893	0.431	0.419
R2 between	0.968	0.964	0.167	0.143

Comment #2

- Clarify the mechanism regarding input sourcing from industries covered / not covered by the ETS

  $\longrightarrow$ Specific focus on this question?

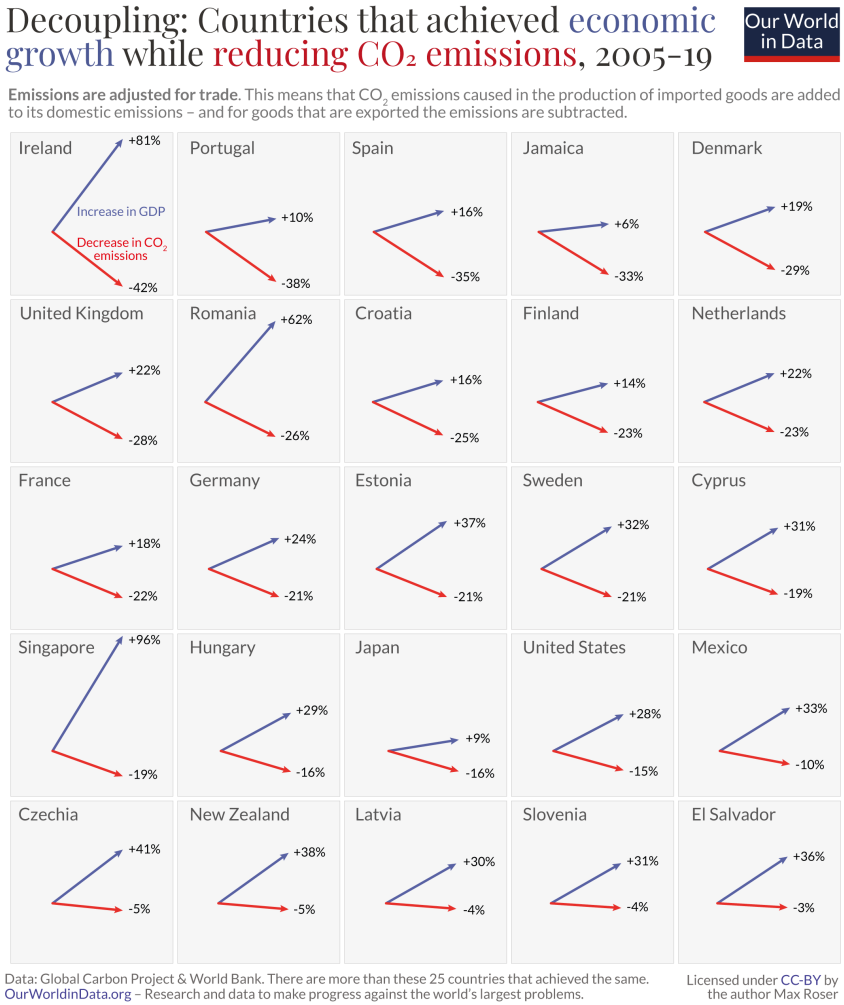
Table (3) Gross output as a function of exposure to ETS regulation by loca

	(1)	(2)	(3)	(4)	(5)	(6)
			input share domestic		input share foreign	
	emisugo	inpsugo	EU_ETS	nonEU_ETS	EU_ETS	nonEU_ETS
non-EU-D	-0.00130***	0.000227	0.00825	0.0120	0.0269	-0.0333
	(0.000361)	(0.00379)	(0.118)	(0.0103)	(0.244)	(0.0236)
non-EU-F	-0.000233	0.00116	0.0690	0.00386	-0.498***	0.0205
	(0.000149)	(0.00272)	(0.0551)	(0.00604)	(0.160)	(0.0325)
EU-D	-0.000719***	0.00371	0.00918	0.0446***	-0.0524***	0.0224
	(0.000121)	(0.00231)	(0.00587)	(0.0138)	(0.00826)	(0.0341)
EU-F	-0.00204***	0.00254	0.00717*	0.00244	-0.0171	0.0260
	(0.000202)	(0.00206)	(0.00425)	(0.00823)	(0.0166)	(0.0216)

Broader question #1: Free allocations

- This paper mobilizes estimations on ETS to give insight on CBAM implementation
- *Free allocations* – one key change that would arise with the implementation of the CBAM :
  - Set in place with the introduction of EU ETS, in order to compensate anti-competitive effects on specific European firms/industries. It raises competition issues *within* the EU
  - To be phased out with the CBAM implementation to avoid advantages for EU-based firms
- A specific focus on the role of free allocations might be very interesting

Broader question #2: Decoupling



- This papers interprets negative correlation between emissions and gross output as an anti-competitive effect
- Broader topic of decoupling between:
  - Emissions adjusted for trade
  - Economic growth
- Room to study this decoupling at a finer level: industry/firm level